

Middle Layers Know Where to Click: Coordinate Serialization Bottlenecks in GUI Agents

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Abstract

Autoregressive GUI agents ground instructions by generating screen coordinates, conflating two computations that the final-layer reading leaves undistinguished: locating the target widget in screen space, and serializing that location into coordinate tokens. We show these computations peak at different transformer depths: spatial grounding is best decoded from intermediate layers and often degrades near the output, while coordinate-token predictability keeps improving in later layers. We call this mismatch the **coordinate serialization bottleneck**. To exploit it, we introduce **ZWERGE-UI**, a frozen-backbone retrofit that reads intermediate-layer posteriors through lightweight cross-attention probes ($\approx 42\text{M}$ parameters, 0.5% of an 8B backbone), fuses them with a $< 200\text{K}$ -parameter head, and routes each sample between the fused point and the backbone’s own coordinate via a training-free confidence gate. Experiments across 4 GUI agents, 2 backbone families, and 5 benchmarks show that the gate yields small but consistent gains under the identical strict success-rate protocol at the cost of a single forward pass.

Introduction

Visual GUI agents are typically trained to ground natural-language instructions by generating screen coordinates or action protocol tokens (Qin et al. 2025; Wu et al. 2025c; Ye et al. 2025; Gu et al. 2025). This end-to-end formulation conflates two computations: *locating* the target widget in screen space and *serializing* that location into a coordinate or action vocabulary. Since the final action is decoded from the last representation, GUI grounding is often analyzed only through the final generated coordinates, leaving unclear whether spatial localization and coordinate serialization emerge at the same depth inside the model.

Prior work has exploited attention maps and prefill-stage signals to aid localization (Xu et al. 2025; Wu et al. 2025a; Lin et al. 2026), while analyses of vision transformers show that final-layer representations may lose spatial fidelity (Darcet et al. 2024; Lan et al. 2024; Chen et al. 2025). However, these lines of work do not ask where *within* the prefill computation spatial grounding is most accessible in

autoregressive GUI agents. At the same time, GUI-agent training studies reveal a tension between chain-of-thought reasoning and precise localization (Xu et al. 2026; Yang et al. 2026; Zhou et al. 2025; Zhang, Gao, and Xu 2025), motivating perception–reasoning decoupling (Zhang et al. 2026). We operationalize this perspective by freezing the backbone and reading spatial beliefs directly from intermediate activations.

We study this question by probing intermediate layers of representative GUI agents with lightweight frozen-backbone readouts. Our analysis reveals a consistent non-monotonic trajectory: spatial grounding improves through early and middle layers, peaks at intermediate depths, and then often degrades toward the final layer. In contrast, a complementary *serialization lens*, teacher-forced coordinate-token likelihood without additional parameters, shows that coordinate tokens become most predictable in later layers. This mismatch suggests that GUI agents first form an internal spatial intent and only later translate it into output tokens. We call this phenomenon the **coordinate serialization bottleneck**.

This bottleneck is not a generic saliency effect. In an instruction-switch counterfactual, changing only the instruction on the same screen shifts the mid-layer spatial posterior from one target to another, indicating that the posterior is instruction-conditioned. The effect is also most pronounced in high-resolution layouts with small targets, where precise screen-space localization is most critical. Together, these findings suggest the final representation is not always the best place to decode GUI grounding: late layers increasingly specialize in expressing coordinates, while intermediate layers preserve more spatially discriminative information.

Motivated by this observation, we introduce ZWERGE-UI, a frozen-backbone retrofit for existing GUI agents. ZWERGE-UI attaches lightweight cross-attention probes to intermediate layers to recover layerwise spatial posteriors, combines them with a small cross-layer fusion head, and then applies a training-free confidence gate that decides, per sample, when to trust the resulting intermediate-layer point over the backbone’s own autoregressive coordinate. The probes serve a dual role: they provide the analytical lens used to reveal the bottleneck, and they form the grounding component used at inference. Importantly, ZWERGE-UI does not modify the agent backbone or its reasoning pathway.

We evaluate ZWERGE-UI on four GUI agents span-

ning two backbone families, Qwen2.5-VL (Qwen Team 2025a) and Qwen3-VL (Qwen Team 2025b), across five GUI grounding benchmarks, ScreenSpot-Pro (Li et al. 2025), ScreenSpot-v2 (Wu et al. 2025b), OSWorld-G (Xie et al. 2025), MMBench-GUI (Wang et al. 2025), and UI-Vision (Nayak et al. 2025), six configurations in total since we evaluate OSWorld-G on both its refusal-removed and original splits. Across models and datasets, the strongest spatial readout consistently appears before the final layer. A training-free confidence gate converts this finding into consistent grounding gains across all four backbones and six benchmarks (§).

In summary, our contributions are as follows:

- We identify the **coordinate serialization bottleneck** in GUI agents: intermediate layers encode more accurate instruction-conditioned spatial posteriors than final layers, while later layers better predict coordinate/action tokens.
- We introduce two complementary diagnostic lenses, a frozen-backbone *spatial lens* and a parameter-free *serialization lens*, and use them to show a robust depth mismatch between spatial grounding and coordinate serialization across four GUI agents.
- We present ZWERGE-UI, a lightweight frozen-backbone retrofit that recovers intermediate-layer spatial signals through cross-attention probes and a training-free, GT-free confidence gate that routes each sample between the fusion point and the native autoregressive coordinate. Across two backbone families and six benchmarks, all four retrofitted agents improve over their native baselines under the identical strict success-rate protocol, at near-zero added cost.

Per-Layer Spatial Probes

We diagnose *where* GUI grounding signals reside across transformer depth by attaching one lightweight probe per layer to a frozen backbone. The same probe set later forms the readout head of the retrofit (§). Probes introduce no new spatial knowledge: the backbone is frozen, and the gradient from layer ℓ reaches only ϕ_ℓ .

Prefill-forced inference. Coordinate generation serializes a location into up to eight number tokens, but the spatial intent is already encoded before decoding begins (§). We prefill a fixed template ending in an anchor token, `click(<|ground|><|pointer_start|>...)`, and run a single forward pass. At the anchor position the model has processed the full screenshot and instruction but no coordinate value, so there is zero label leakage.

Cross-attention probe. Let f_θ be a frozen L -layer backbone. For each probed layer $\ell \in \mathcal{L}_{\text{probe}}$, drawn from the deeper half of the network where spatial signal concentrates, we read the anchor query $\mathbf{h}_{t^*}^{(\ell)} \in \mathbb{R}^d$ at the `<|ground|>` position and visual keys $\mathbf{H}_v^{(\ell)} \in \mathbb{R}^{N_v \times d}$ over the N_v patch tokens. Each probe ϕ_ℓ computes a patch-level posterior via

multi-head cross-attention with a learnable head gate α :

$$\begin{aligned} \mathbf{Q}_\ell &= W_q^{(\ell)} \text{Norm}(\mathbf{h}_{t^*}^{(\ell)}), \quad \mathbf{K}_\ell = W_k^{(\ell)} \text{Norm}(\mathbf{H}_v^{(\ell)}), \\ p_\ell &= \sigma \left(\sum_h \sigma(\alpha)_h \frac{\mathbf{Q}_{\ell,h}^\top \mathbf{K}_{\ell,h}}{\sqrt{d_h}} \right) \in \Delta^{N_v-1}, \end{aligned} \quad (1)$$

where $\text{Norm}(\cdot)$ is the RMS pre-scale followed by LayerNorm, and $W_q^{(\ell)}, W_k^{(\ell)}$ are full-rank projections so each probe learns a geometry suited to its layer. Each probe is $\approx 4.2\text{M}$ parameters ($d=4096, n_h=8, d_h=64$); the full set totals $\approx 42\text{M}$, or 0.52% of a 7B backbone.

RMSNorm pre-scale. Hidden-state ℓ_2 norms grow with depth, geometrically misaligning layers. We pre-scale $\mathbf{h} \leftarrow \mathbf{h}/(\|\mathbf{h}\|_2/\sqrt{d})$ before each LayerNorm, bounding the input norm to $\approx \sqrt{d}$ regardless of depth and preventing `bfloat16` overflow.

Supervision. Each probe is supervised with an anisotropic Gaussian centered at the bbox centroid:

$$y_i \propto \exp \left(-\frac{(x_i - c_x)^2}{2\sigma_x^2} - \frac{(y_i - c_y)^2}{2\sigma_y^2} \right), \quad \sigma_x = \eta w_b, \sigma_y = \eta h_b \quad (2)$$

which reduces over-penalization from patch-grid quantization on small targets (§). Stage 1 minimizes $\mathcal{L}_{S1} = \frac{1}{|\mathcal{L}_{\text{probe}}|} \sum_\ell D_{\text{KL}}(\mathbf{y} \| p_\ell)$ with the backbone receiving zero gradient. Training-data construction is detailed in the supplementary material.

The Coordinate Serialization Bottleneck

We investigate *where* GUI grounding signals are encoded across transformer layers, building toward four findings that motivate our method.

Experimental setup. We study four GUI agents spanning two backbone families, Qwen2.5-VL (Qwen Team 2025a) and Qwen3-VL (Qwen Team 2025b), on ScreenSpot-Pro (Li et al. 2025) and ScreenSpot-v2 (Wu et al. 2025b). Spatial grounding quality is measured via a lightweight per-layer grounding head, Stage 1 of §, which is about 0.5% of backbone parameters and is trained with the backbone completely frozen, so it introduces no new spatial knowledge. Coordinate serialization quality is measured via the backbone’s native language model head under teacher forcing, with no additional parameters.

Spatial Grounding Peaks at Intermediate Layers

Autoregressive GUI agents implicitly assume final-layer representations carry the richest spatial information. The blue curves in Figure 1 challenge this: the spatial signal peaks at intermediate depths, then declines toward the output. This non-monotonic trajectory is consistent across both backbone families, ruling out any progressive-refinement interpretation.

Finding 1. *Intermediate transformer layers are consistently better spatial decoders than the final layer.*

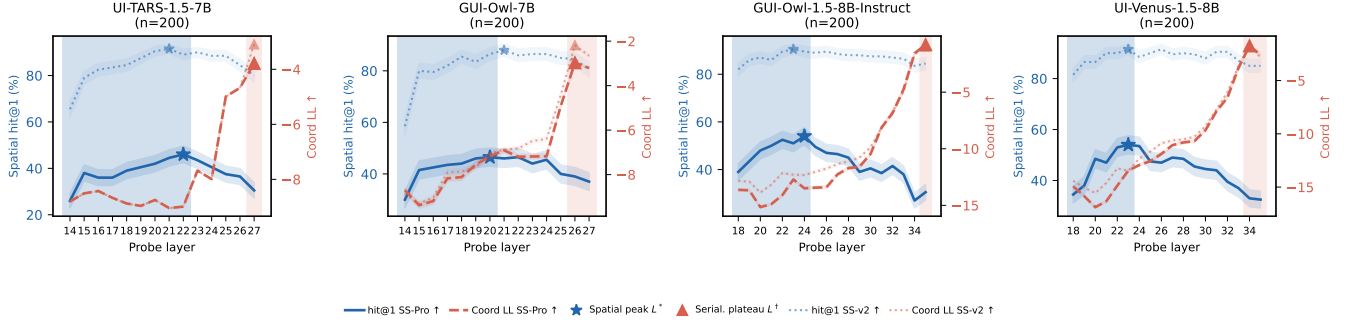


Figure 1: **Spatial localization and coordinate serialization peak at different depths.** Blue curves (left axis): spatial hit@1 $\uparrow$. Orange curves (right axis): coordinate token log-likelihood $\uparrow$. Both signals rise together, then diverge at the spatial peak L^* (blue star): spatial quality declines while serialization continues rising to $L^\dagger$ (orange triangle). The consistent lag $L^\dagger > L^*$ across all models constitutes direct evidence for the coordinate serialization bottleneck.

Robustness to UI-layout distribution and sample size.

The inverted-U is not an artifact of checkpoint overfitting or the modest $n=200$ lens sample. Across all five benchmarks, including the distribution-shifted UI-Vision and MMBench-GUI, the spatial-peak layer L^* varies by at most 3 layers per model with a standard deviation of at most 1.1, reported in full in the supplementary material, and peak exceeds final in 20/20 cells. Bootstrap resampling with 1000 resamples, also reported in the supplementary material, places $L^\dagger$ within a ≤ 1 -layer 95% confidence interval for every model and L^* within ≤ 3 layers for three of four models, and ≤ 7 layers for the noisiest model, GUI-Owl-7B, with $L^* < L^\dagger$ holding in all four models. Absolute hit@1 carries a ± 2 –4 pp 95% confidence interval, so we claim the *shape* of the trajectory, not the absolute values.

A Dissociation Between Spatial Localization and Coordinate Serialization

If late layers are not spatial refiners, what do they do? We hypothesize they are dedicated to *coordinate serialization*, translating a latent spatial intent into the vocabulary-level token sequence required by the autoregressive head. We test this with a *serialization lens*: coordinate token log-likelihood measured via teacher-forced inference at each layer.

The red curves in Figure 1 reveal a striking dissociation. Both signals rise together through early layers, then diverge: spatial quality peaks and falls, while coordinate log-likelihood continues climbing to a later plateau. The lag between the spatial peak and the serialization plateau is consistent across all four models and both backbone families, providing direct evidence for a **coordinate serialization bottleneck**.

Finding 2. *The layer that best decodes the target region consistently precedes the layer that best predicts the coordinate token sequence. Late layers are coordinate serializers, not spatial refiners.*

Logit-lens controls. Applying the native LM head to intermediate states is a logit-lens-style probe (nostalgebraist 2020). A skeptical reading is that late layers could simply align better with the output embedding space for reasons unrelated to coordinates, rather than because they specifically refine coordinate content. Layerwise probing results in general can be artifacts of where a frozen readout head happens to align best (Geva et al. 2021; Meng et al. 2022), so we run two controls to isolate a coordinate-specific effect from this generic alignment confound; full tables are in the supplementary material.

Non-coordinate control. We repeat the same teacher-forced measurement on *non-coordinate* protocol and format tokens drawn from the same generations, such as parentheses, quotation marks, and JSON keys. These reach near-zero NLL at late layers for three of four models, for example UI-TARS-1.5-7B drops from 15.4 to 0.0 and GUI-Owl-1.5-8B drops from 7.9 to 0.0, confirming that late layers do align generically with the output vocabulary. GUI-Owl-7B is a partial exception: its protocol NLL rises again toward the output, from 8.2 to 12.4. Crucially, the coordinate *value* tokens nonetheless retain a substantial residual NLL at each model’s plateau, 2.2–3.1 nats for the two Qwen2.5-VL models and 1.0–1.7 nats for the two Qwen3-VL models. Even where format tokens are fully serialized by the final layers, the numeric value is not, and this residual is the coordinate-specific component of the bottleneck rather than a generic alignment artifact. The plateau layer $L^\dagger$ itself coincides for coordinate and protocol tokens in 3 of 4 models, differing by only 2 layers for GUI-Owl-7B, so $L^\dagger$ ’s *location* is a general late-layer phenomenon while the coordinate-specific signal is the residual NLL gap that survives at that layer.

Tuned-lens control. Following the tuned-lens methodology (Belrose et al. 2023), we fit a lightweight per-layer affine map from each intermediate state to the final-layer representation and apply the *frozen* output head to the calibrated state. This flattens the raw depth trend: early-layer NLL drops from 8.6 to 3.8 nats after calibration, conceding

that part of the raw improvement with depth is indeed an alignment artifact. Yet the plateau layer itself is unchanged at L26 both before and after calibration, so the *locus* of the serialization plateau is robust to alignment calibration even though its raw slope is partly confounded. Together, the residual coordinate-specific NLL gap and the calibration-invariant plateau location indicate that the observed lag reflects genuine coordinate serialization rather than a probing artifact.

The Mid-Layer Posterior Is Instruction-Conditioned, Not Visual Saliency

A skeptical reading is that mid-layer spatial signals merely reflect *visual saliency*: prominent targets appear highlighted regardless of the instruction. We design a counterfactual to falsify this: for pairs of records sharing the same screen but targeting disjoint elements, we swap the instruction while keeping the image fixed and measure how much the spatial posterior shifts.

Figure 2 shows the posterior shifts dramatically. Switching the instruction collapses the spatial mass on the original target by a substantial margin, concentrated at exactly the intermediate layers where grounding quality peaks. This decisively rules out the saliency account: the mid-layer representation is an instruction-conditioned spatial posterior, not a passive saliency map.

Finding 3. *Swapping the instruction while keeping the image fixed causes a large shift in the mid-layer spatial posterior. The spatial signal is instruction-conditioned grounding, not visual saliency.*

The Bottleneck Is Amplified on High-Resolution Small-Target Layouts

We ask whether the bottleneck is *amplified* in the setting where GUI grounding is hardest: high-resolution screen-shots with small, densely packed targets.

We quantify severity via the *collapse gap*: peak-layer hit@1 minus final-layer hit@1. Table 1 shows the gap is consistently larger on the high-resolution, small-target ScreenSpot-Pro than on the mid-resolution, larger-target ScreenSpot-v2, across every model. This is not a difficulty artifact: even though absolute performance is higher on SS-v2, the relative degradation from peak to final layer is smaller. The serialization overhead is most costly precisely where spatial precision matters most.

Controlled within-benchmark target-size regression. A concern with Table 1 is that SS-Pro vs SS-v2 conflates target size with resolution and UI density. We hold the benchmark fixed at ScreenSpot-Pro, with $n=1581$, and bin samples by target bbox area, reported in full in the supplementary material. The peak-final *gap* does *not* increase monotonically as targets shrink: it rises monotonically with size for the two Qwen2.5-7B models, from 6–10 to 20–22 pp, but peaks at intermediate sizes and falls back for the two Qwen3-8B models, 22–25 pp at the third quintile versus 14–18 pp at the fifth. Averaged over models the trend is non-monotone, with

Model	SS-Pro			SS-v2		
	Pk	Fin	Gap↓	Pk	Fin	Gap↓
UI-TARS-1.5-7B	46.0	30.5	15.5	91.5	80.0	11.5
GUI-Owl-7B	46.5	37.0	9.5	88.0	80.5	7.5
GUI-Owl-1.5-8B	54.0	30.5	23.5	90.5	84.5	6.0
UI-Venus-1.5-8B	54.0	32.5	21.5	91.5	85.0	6.5

Table 1: **Collapse gap is larger on high-resolution layouts.** Pk = peak-layer hit@1; Fin = final-layer hit@1 (pp). SS-Pro gaps consistently exceed SS-v2 gaps.

the five quintiles reading 13.0, 16.3, 20.9, 18.5, and 18.6 pp, mechanically floored by peak-layer accuracy on sub-patch targets where both peak and final fail. The gap is thus not a monotone target-size effect. *However*, absolute final-layer hit@1 is about 34% on the smallest quintile versus about 58% on the largest, roughly $1.7\times$ worse and ranging from $1.5\times$ to $1.9\times$ per model, so the bottleneck is most damaging in *absolute* terms where spatial precision matters most. This is the claim Finding 4 makes via absolute accuracy rather than the confounded gap.

Finding 4. *The collapse gap from peak to final layer is consistently larger on high-resolution benchmarks with small targets. The coordinate serialization bottleneck is most damaging precisely where spatial precision matters most.*

Taken together, the four findings motivate reading the spatial posterior from intermediate layers and using cross-layer fusion to select the optimal depth per sample, as described below.

ZWERGE-UI

The per-layer probes of § are independent, and the best layer varies by sample (§, Finding 1). We add two lightweight components on top of the frozen probes, shown in Figure 3: a cross-layer fusion head that selects the reliable depth per sample, and a training-free confidence gate that decides when to trust the fused point over the backbone’s own decoder. Both reuse the single prefill of §, so the retrofit adds negligible compute over the native baseline.

Cross-Layer Fusion

We draw active layers $\mathcal{L}_{\text{active}} \subset \mathcal{L}_{\text{probe}}$ from the spatial plateau identified in § and fuse their posteriors. A cross-layer LoRA refines each anchor query $\mathbf{h}_{t*}^{(\ell)}$ into $\tilde{\mathbf{q}}_{\ell}$; a cosine-meta scorer then assigns per-layer weights:

$$\omega_{\ell} = \text{softmax}(\alpha_{\ell} + \tau^{-1} \cos(\tilde{\mathbf{q}}_{\ell}, \mathbf{q}_{\text{meta}})), \quad (3)$$

$$p_{\text{final}} = \sum_{\ell \in \mathcal{L}_{\text{active}}} \omega_{\ell} p_{\ell},$$

where α_{ℓ} is a sample-independent trainable prior on layer ℓ ’s average quality, $\mathbf{q}_{\text{meta}}$ is a global trainable meta-query serving as a soft prototype of the ideal grounding representation, and τ is a learnable temperature. Layers whose refined

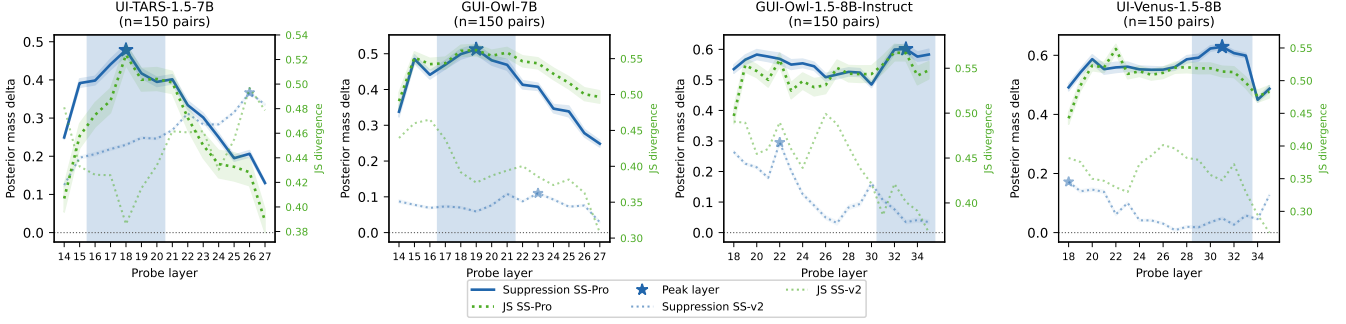


Figure 2: **Instruction-switch counterfactual: the mid-layer posterior is instruction-conditioned.** Blue curves: old-target suppression $\uparrow$. Orange curves: posterior JS divergence $\uparrow$. Both effects peak at the same intermediate layers as spatial grounding quality (cf. Fig. 1) and are substantially nonzero, ruling out visual saliency.

query aligns with $\mathbf{q}_{\text{meta}}$ receive higher weight for the current input. The LoRA contributes $\approx 200\text{K}$ parameters, two orders of magnitude below a single probe.

Stage 2 loads the Stage-1 probes; the backbone and inactive probes stay frozen, while the fusion head and the active probes remain trainable, with loss

$$\mathcal{L}_{S2} = D_{\text{KL}}(\mathbf{y} \parallel p_{\text{final}}) + \frac{\lambda}{|\mathcal{L}_{\text{active}}|} \sum_{\ell} D_{\text{KL}}(\mathbf{y} \parallel p_{\ell}), \quad \lambda = 0.2, \quad (4)$$

so the per-layer auxiliary term gives every active probe a direct gradient signal and no single layer is neglected.

Training-Free Gated Decoding

The fusion posterior p_{final} is patch-level, not a coordinate. In one prefill we read both the fused point $\mathbf{f}$, produced by a training-free posterior-to-point decoder detailed in the supplementary material, and the backbone’s native autoregressive coordinate $\mathbf{n}$, and route between them with a GT-free gate on the posterior sharpness: emit $\mathbf{f}$ when the margin $m = p_{\text{final}}^{(1)} - p_{\text{final}}^{(2)}$, the difference between the top-two single-patch probabilities, exceeds τ , otherwise emit $\mathbf{n}$. A sharp mode means the probe is confident in one location; a diffuse mode means the native serializer should be trusted. The default is $\tau=0.25$, while the main results use 0.20. This instantiates the paper’s thesis: the intermediate layer identifies *where* the target is, and a confidence gate decides when to trust that signal over the final-layer serializer.

Fairness and cost. The comparison against the native baseline is protocol-symmetric: both points are evaluated under the same strict point-in-box metric on the same cached run, and the gate uses no labels. The ensemble adds no autoregressive generation over the baseline. The native generate is the baseline’s own computation, and the fused point reads intermediate activations that the backbone already produces in that same forward; in principle the two share one prefill, so the full retrofit is realizable at the cost of a single forward pass.

Experiments

Experimental Setup

Benchmarks. We evaluate on six GUI grounding configurations: ScreenSpot-Pro (Li et al. 2025), ScreenSpot-v2 (Wu et al. 2025b), the original and refusal-removed OSWorld-G-refine splits of OSWorld-G (Xie et al. 2025), MMBench-GUI (Wang et al. 2025), and UI-Vision (Nayak et al. 2025), spanning desktop, web, and mobile.

Metrics. We report the strict single-point success rate (overlap@1); protocol and patch-mapping details are in the supplementary material.

Baselines. We retrofit four GUI agents: GUI-Owl-7B (Ye et al. 2025) and UI-TARS-1.5-7B (Qin et al. 2025), both built on the 28-layer Qwen2.5-VL, and GUI-Owl-1.5-8B (Xu et al. 2026) and UI-Venus-1.5-8B (Team et al. 2026), both built on the 36-layer Qwen3-VL. Each backbone is frozen throughout; only the probe and fusion heads are updated. The Native column is each backbone’s own autoregressive accuracy on the same task set, so the comparison isolates the retrofit.

Implementation. Stage 1 trains 14–18 CrossAttn probes per model on 200k grounding samples for one epoch; Stage 2 trains the fusion head with the active probes (12 active layers) for one epoch, backbone and inactive probes frozen. The deployed decoder is the margin-gated ensemble (§) with $\tau=0.20$.

Main Results

Table 2 compares the confidence-gated ensemble against the native baseline under the identical strict success-rate protocol. Across two backbone families and six benchmarks, all four retrofitted agents improve over their native baselines, with a single global threshold and no per-backbone tuning. The gate routes to the fusion point on 5.7–32.7% of samples and otherwise defers to the native decoder; it fires most on OSWorld-G, where the native baseline has the most headroom, and least on ScreenSpot-v2, where the native baseline is already near ceiling.

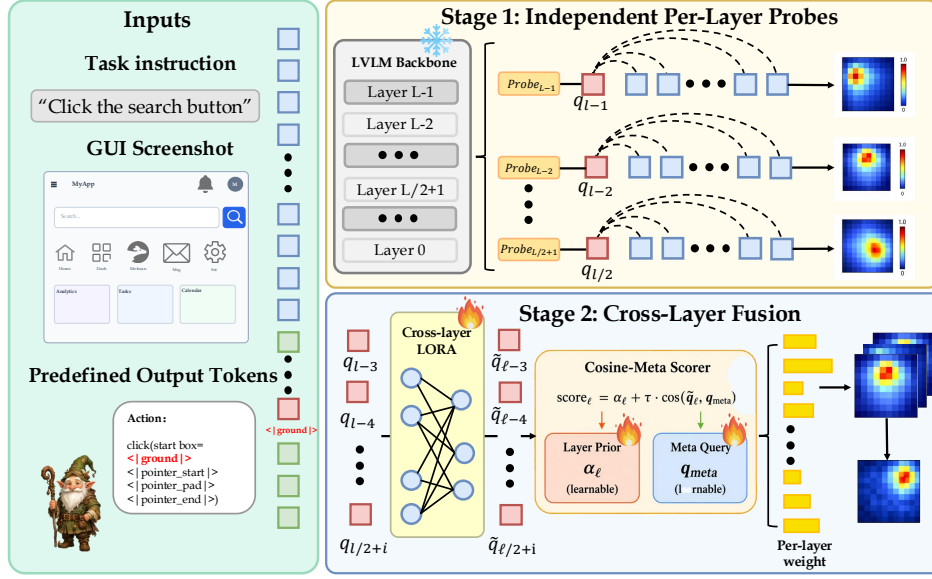


Figure 3: **Overview of ZWERGE-UI.** A frozen backbone exposes layerwise hidden states from a single prefill; independent probes decode intermediate-layer spatial posteriors, a cross-layer fusion head selects the reliable depth per sample, and a training-free confidence gate routes each sample between the fused point and the native autoregressive coordinate.

Modest gains, by design. Our aim is not to top leaderboards but to validate the coordinate serialization bottleneck: a near-zero-cost, GT-free gate that trusts the intermediate-layer posterior when it is sharp consistently improves grounding, confirming that the spatial signal resides before the final layer. The probe and fusion heads train on only 200k samples, a small fraction of the backbone’s pre-training budget, so these gains are a lower bound. The ensemble adds no autoregressive generation over the baseline and is realizable at the cost of a single forward pass, as described in §.

Diagnosing the Gains: Target Geometry and Retrofit Headroom

Table 2 shows the largest absolute headroom on ScreenSpot-Pro and the smallest gains on Qwen3-VL backbones. We explain both patterns below with two lightweight diagnostics computed from the same cached runs, at no extra inference cost.

Why ScreenSpot-Pro is the hard case. We measure the smaller side of every ground-truth bounding box across all 1,581 ScreenSpot-Pro targets and compare it against one visual-patch width, shown in Figure 4. Roughly three quarters of targets are smaller than a single patch, so a patch-level posterior can at best localize the correct patch, not the sub-patch pixel. Only the backbone’s own autoregressive decoder, which regresses a continuous coordinate rather than a patch index, can resolve position within that patch. This is precisely why the retrofit is framed as a *gate* rather than a replacement decoder: the intermediate posterior identifies *which patch* contains the target, while sub-patch precision is left to the native serializer whenever the probe is uncertain

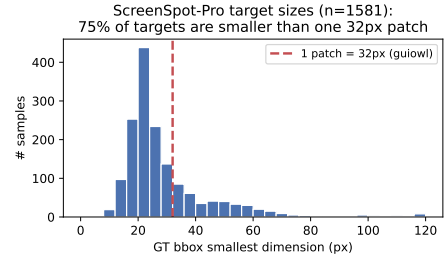


Figure 4: **Three quarters of ScreenSpot-Pro targets are smaller than one visual patch.** Distribution of the smaller GT-bbox dimension over all 1,581 ScreenSpot-Pro targets; the dashed line marks one patch width.

which patch to trust.

Retrofit headroom predicts the gate’s behavior. For each backbone and benchmark, define the *retrofit headroom* $h = \max_{\ell} \text{ov}@1(p_{\ell}) - \text{ov}@1(\text{native})$, the gap between the best intermediate-layer probe and the native baseline. When $h > 0$, bypassing the serializer recovers real headroom; when $h \leq 0$, the native decoder has already closed that gap. Figure 5 plots h against the strict, ungated fusion-only delta. Qwen2.5-VL backbones, whose native decoders leave more headroom on the table, occupy the positive-headroom, positive-gain quadrant, while Qwen3-VL backbones, whose native decoders already approach the intermediate-layer ceiling, sit near or below zero on both axes. This is exactly the regime the margin gate of Eq. 3 targets: it defers to the native decoder whenever headroom is low, explaining why a single global threshold makes every cell in Table 2 non-negative.

Table 2: **Confidence-gated ensemble, strict success rate (overlap@1, %).** **Native:** backbone autoregressive baseline. **+ZwerGe:** margin-gated fusion-native ensemble (§), identical protocol, same cached run. $\Delta = +\text{ZwerGe} - \text{Native}$. **%Fus.:** fraction of samples the gate routes to the fusion point. OSW-G = OSWorld-G (original); OSW-G-Ref = OSWorld-G-refine (refusal-removed). All four backbones improve on average across the six benchmarks (macro-mean $\Delta = +0.46$ pp), with one global threshold $\tau=0.20$ and no per-backbone tuning.

Backbone	Benchmark	Native ov@1	+ZwerGe ov@1	Δ	%Fus.
GUI-Owl-7B (Ye et al. 2025) Qwen2.5-VL	SS-Pro	55.22	56.10	+0.89	7.5
	SS-v2	92.84	93.08	+0.24	5.8
	OSW-G-Ref	79.61	80.39	+0.78	25.7
	OSW-G	72.55	72.94	+0.39	25.3
	MMBench-GUI	80.77	81.27	+0.50	7.5
	UI-Vision	53.87	54.42	+0.55	12.7
UI-TARS-1.5-7B (Qin et al. 2025) Qwen2.5-VL	SS-Pro	48.32	49.59	+1.27	6.8
	SS-v2	91.35	91.66	+0.31	5.7
	OSW-G-Ref	77.06	76.47	−0.59	24.5
	OSW-G	64.51	65.10	+0.59	20.8
	MMBench-GUI	77.46	77.88	+0.42	7.6
	UI-Vision	27.12	28.45	+1.33	7.6
GUI-Owl-1.5-8B (Xu et al. 2026) Qwen3-VL	SS-Pro	74.89	75.21	+0.32	10.2
	SS-v2	94.10	94.02	−0.08	9.9
	OSW-G-Ref	80.98	82.35	+1.37	32.7
	OSW-G	78.43	78.63	+0.20	30.4
	MMBench-GUI	83.58	84.17	+0.58	9.9
	UI-Vision	43.94	44.97	+1.04	14.2
UI-Venus-1.5-8B (Team et al. 2026) Qwen3-VL	SS-Pro	70.52	70.65	+0.13	7.7
	SS-v2	96.85	96.85	0.00	8.6
	OSW-G-Ref	86.08	86.27	+0.20	28.8
	OSW-G	81.76	82.35	+0.59	25.5
	MMBench-GUI	89.04	88.95	−0.08	8.8
	UI-Vision	52.18	52.33	+0.16	11.8

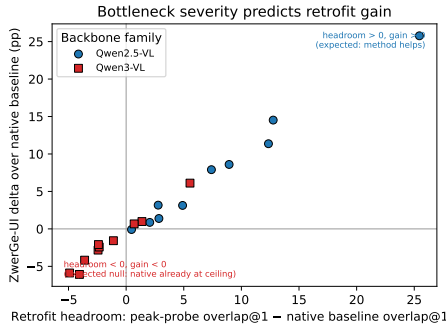


Figure 5: **Retrofit headroom predicts the ungated fusion-only gain.** x -axis: headroom $h = \max_{\ell} \text{ov}@1(p_{\ell}) - \text{ov}@1(\text{native})$. y -axis: strict, ungated fusion-only overlap@1 delta.

Conclusion

We presented ZWERGE-UI, a frozen-backbone retrofit that exposes and exploits the coordinate serialization bottleneck in GUI agents: instruction-conditioned spatial posteriors peak at intermediate transformer layers, while later layers increasingly specialize in coordinate serialization rather

than localization refinement. A training-free, GT-free confidence gate combines the fused intermediate-layer point with the native decoder at the cost of a single forward pass. Across two backbone families and six benchmarks, all four retrofitted agents improve over their native baselines. The gains are modest by design: the retrofit’s purpose is to validate the bottleneck hypothesis rather than chase leaderboard scores, and the heads train on far less data than the backbone, leaving clear room for improvement as GUI agents scale.

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